

THE WORLD BEFORE THE WORD



The river, the deer, and the architecture of answerability

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The river

Tell someone where to find water.

Down the valley, past the burnt tree, to the bend where the current slows. They walk. Some time later, they drink.

Nothing about this is remarkable. That is why it has never been examined properly.

Examine it.

The river is there or it is not. The person walks to the bend and finds water, or finds a dry bed and comes back thirsty. The words were answerable to something outside themselves, and the world settled which way it went.

Not the fluency of the description. Not its internal coherence. Not whether it sounded like the sort of thing a person who knew the country would say.

The water.

Answerability

This is the primitive act, and everything else in language is built on it or borrowed from it.

An utterance reaches past itself to a domain that does not depend on the utterance, and that domain decides the matter.

Note what is being claimed and what is not. Not that every meaningful expression has an existing referent — it does not, and the objections are old and correct. Not that speech becomes impossible when reference fails. *The river has dried up* is perfectly meaningful. *Go to Atlantis* is a performable instruction. What fails in these cases is the drinking.

The act survives. The consequence does not follow.

So the requirement is not a settled referent. It is a standing domain against which the claim can be assessed — and a determinate standing within it, which may be positive, negative, or explicitly unresolved. *There is no water at the bend* is a standing. *Whether this bend is that bend has not been established* is a standing. Absence and uncertainty are conditions of the frame, not gaps in it.

The frame holds standing. It does not require the entity.

What must stand

Four things, and none of them is the referent.

A domain that persists. Something the claim ranges over, continuing independently of the claim and of anyone attending to it. Without this the words range over nothing and no observation could bear on them.

Criteria of individuation that hold. What counts as the same river, stable across the walk. Not similarity — criteria. A description that picks out whatever currently best fits it has not picked out anything; it has described a neighbourhood.

An interval across which the claim binds. Words spoken here govern an arrival there, with no words present at the moment of drinking. Whoever acts must persist to be the one who acted. And the claim carries a time at which it is answerable — a direction may be true when spoken and stale when acted upon, which is a different failure from having been false.

An attribution that persists. The claim answers *for* something. Not necessarily for a living speaker: directions outlive their authors, and hearsay, inscriptions and centuries-old instructions remain action-guiding without anyone left to hold responsible. What must persist is the attribution itself — source, basis, scope, and whatever standing that source presently has. The record endures while the authority of its source may lapse independently, and those two facts must be held apart rather than collapsed.

Remove any one and the act does not become impossible. It becomes unable to succeed on its own terms. The words still issue. The walk still happens. The drinking is what fails.

A deer in Prato

Consider the hardest case, which is negation.

There is no water at the bend presupposes the bend, and the presupposition does the work. But negation does not generally require an entity bearing a failed property. *There is no largest prime. Nobody entered the room. Unicorns do not exist.*

Take the last one, because it is the instructive one.

I cannot prove that unicorns never existed. The claim quantifies over every entity at every time, and finite evidence does not reach that far. What the claim requires is a domain over which it ranges, criteria determining what would count as a unicorn, identity conditions distinguishing one from other animals, and evidence proportionate to the confidence asserted.

The proposition is meaningful because it is answerable to a domain. No member of that domain need satisfy the predicate.

Which leaves the universal negative permanently exposed. A single instance falsifies it. No quantity of absence establishes it. The asymmetry does not diminish with effort.

In June 2008 the Associated Press reported a roe deer, a year old, born in captivity at the Center of Natural Sciences in Prato near Florence, with a single horn in the centre of its head. Park officials nicknamed him Unicorn and attributed the feature to a developmental anomaly.

His twin has two horns.

The twin is what makes this a demonstration rather than a curiosity. Same dam, same litter, same reserve, same day, same criteria available to be applied to both. Roe deer twins are usually dizygotic, so these are siblings rather than copies — which is beside the point, since nothing here turns on their genetics. Two animals were produced under identical conditions of observation, and only one of them makes anyone reach for a word out of mythology.

So state the criteria. If unicornhood is a single central horn, this animal satisfies it and the matter is closed. If unicornhood requires the white horse, the spiral, the healing powers and the maidens, then the criteria were never about horns, and everyone asserting the universal negative had been carrying a specification they never declared.

Either way, the confident negative was not running on evidence. It was running on an unstated specification applied to an unbounded domain, and it stood for centuries because nothing had ever turned up to test it.

One animal, two criteria

Look at him directly.

A male roe deer stands there bearing a single central projection — the feature by which the witnesses recognised him, and by which they named him.

Zoology classifies that structure as an anomalous antler. It grew from a pedicle, it is bone, it is cast and regrown seasonally, and it therefore belongs to a category that has nothing to do with horns. Ordinary language, and the people standing in the enclosure looking at him, classify it as a horn. One projection, central, forward, on a head.

The animal is unchanged throughout. So is the structure. Two systems assign it different standing, and neither of them is making a mistake.

They diverge because they individuate differently and always did. Zoology individuates by ontogeny and material. Ordinary language individuates by what presents. Both criteria are coherent. Both are competently applied. Nobody had said which was in force, because nobody had noticed that the question could be asked.

This is not a quarrel about a word, and the deflationary reading fails on the same evidence. Consequences follow differently under each criterion. Under the zoological one, the correct act is a line in a record of developmental anomalies. Under the ordinary one, the correct act is to name him Unicorn and telephone the Associated Press. Both acts were performed. Neither is derivable from the other without first establishing which criterion governs.

Which is the whole structure, visible in one animal.

The frame holds the deer. The deer persists through the disagreement, unbothered, chewing. What varies is standing under a criterion — and standing is criterion-relative in a way that existence is not.

An entity can be one thing. Its standing can be several, depending on what is in force. A system that cannot hold which criterion governs cannot establish standing at all. It can only report a resemblance and let whoever is reading supply the rest.

How the myth was made

The director of the centre described the animal as fantasy becoming reality, and in the same breath restated that the unicorn has always been a mythological creature. Both at once, with the animal standing in his own park under the name he had given it. The criteria doing the real work were the horse and the magic, and those had never been stated, never been testable, and were never about horns at all.

Then he said the more interesting thing. Anomalies of this kind may be what inspired the myth in the first place.

That reverses the direction of the whole case.

On this reading the mythological unicorn is not a creature that failed to exist. It is a real observation — an animal like this one, seen and reported — that lost its standing in transmission. Each retelling added what the previous telling made plausible. The horn acquired a spiral, the deer acquired a horse, the horse acquired healing, the healing acquired a maiden. By the time it reaches the medieval bestiaries the description is fluent, vivid, richly attributed, and no longer about anything anyone could check.

The referent was real. Nothing held it.

That is drift, and it is worth naming precisely, because the mechanism is not exotic. Each step preserved the surface form of a description while losing the standing of the observation the description came from. No stage was a lie. No stage was even much of an error. What was missing at every stage was anything that carried forward what had actually been established, under what criteria, on whose authority, and within what bounds — so that the next teller was constrained by it.

A thousand years of transmission with no persistent frame produces exactly this: a well-formed description that has quietly stopped being answerable to anything.

Hold that thought. It will return with better funding.

The deer, meanwhile, was specified. One horn, a body, a reserve, a date, a twin. Assessable, and capable of being got wrong.

So the lawful claim is not *unicorns never existed*. It is: no unicorn has been established within the examined domain, under these criteria.

That is not a hedge. It is the true proposition. The confident version manufactures universal certainty out of bounded evidence, and it took an actual deer, with a brother, to expose it.

What this is not

The argument does not run from the conditions of meaningful thought to conclusions about the world. That route yields at most the claim that we must *represent* things as persisting, which is the idealist deflation and I do not want it.

The argument runs through the walk.

People act on descriptions and arrive at water. Strangers act on directions given by the dead and arrive at water. Instructions for making fire cross generations and the fire lights. This works, repeatedly, across people who have never met, and it would be inexplicable if nothing stood on the other end.

That is an inference from what consequence reliably does, not from what language requires us to suppose. The world pushes back, and the pushing back is the evidence.

An argument from what there is. Not an argument for what can be conceived.

The corollary

One consequence, stated now because everything downstream depends on it.

Language can speak beyond settled existence. Fiction, hypothesis, mathematics, mistaken identity, disputed objects, and unresolved reference all operate without a settled referent, and they operate well.

What cannot be done lawfully is to produce consequences as though unsettled existence were settled.

The mention is free. The act is not.

The serious alternative

What follows is a demonstration, not a change of subject.

The argument so far concerns language, and the conditions under which any claim can be answerable to anything at all. Machines enter here because they are where those conditions are currently being violated at scale, with enormous resources, by people working carefully and in good faith. If the account of answerability is correct, a system built without a standing frame should fail in specific and predictable ways — and the fairest place to look is not the weakest architecture but the most serious attempt to move beyond the one everybody already criticises.

Yann LeCun has diagnosed the central weakness of large language models more clearly than most of the industry. A machine trained to continue text does not acquire physical common sense merely by becoming very good at continuing text. Intelligence requires some capacity to

represent the world, predict how it will change, and choose actions by considering their possible consequences.

His proposed route is the [Joint Embedding Predictive Architecture](#). Instead of predicting every pixel or every word, a JEPA learns abstract representations of observed states and predicts the representations of missing or future states. In [V-JEPA 2](#), an encoder maps video into embeddings said to capture useful semantic information about the observed world. A predictor estimates further embeddings. The action-conditioned version receives previous states and possible actions, predicts the representation of the next video frame, and lets a robot choose the action whose predicted result comes closest to a goal image.

This is a real advance. The robot has more than a sequence. It has perception, learned state, counterfactual transition and feedback from action. It can reach for an unfamiliar object, observe what happened, and plan again.

It is still on the wrong side of the river.

A world model is still a model

The phrase *world model* hides the decisive step. What the architecture contains is a learned representation of observations and a function over changes in that representation. The world remains outside it.

Write the action-conditioned operation in its simplest form:

$$s(t+1) = f(s(t), a(t))$$

Given its representation of the present state and a possible action, the model predicts a representation of what comes next.

The function estimates a future state from a present state and an action. It can be operationally useful without individuating a single object. A dynamical model may learn that one field of values tends to follow another and predict the transition extremely well.

The prior question appears as soon as the transition is treated as the change of some continuing thing:

What is it whose state has changed?

The variable s does not supply that answer. It marks a position in the calculation. Similar vectors may belong to different entities. Dissimilar vectors may belong to one entity that has changed radically. An object can disappear behind another and return; two indistinguishable objects can cross; a tool can be repaired until none of its original parts remain; a river can keep its name while its bed moves and its water is replaced continuously.

Nothing prevents the predictor from working through these cases. What it cannot derive from predictive success is whether its successive representations concern the same thing. The moment a predicted transition is used as evidence about *this cup*, as satisfaction of *this goal*,

or as warrant for an action upon *this person*, an identity criterion has entered from somewhere else.

Prediction can learn which continuation is likely in each case. Likelihood cannot establish which criterion of identity governs the case. That criterion determines whether a later observation is an update to the same entity, evidence about another entity, or insufficient to decide.

So object continuity inferred from representational similarity is an estimate. A persistent frame can hold a stronger and more difficult result: these dissimilar observations belong to one continuing entity; these similar observations belong to two; this identity remains unresolved, and no consequence requiring settled identity may proceed.

Recent object-centred variants do not change the logical position. [Causal-JEPA](#), co-authored by LeCun, masks and predicts object-level representations to improve relational and counterfactual reasoning. An object slot is still a representational locus. It does not establish the criteria under which an object at one time is the same object later, nor can the slot make its own unresolved identity binding.

The dependency is visible in the architecture. C-JEPA relies on a video slot encoder designed to maintain slot identity over time, then preserves the earliest object state as an identity anchor across the masked interval. This keeps the predictor's tokens aligned. It does not establish the criterion under which an observed entity remains the same entity, or specify what must follow when that continuity cannot be established. Identity is operationally maintained so prediction can proceed. It is not adjudicated.

[LeWorldModel](#) makes end-to-end latent prediction more stable and much faster for planning. Stability improves the predictor. It does not confer standing on what is predicted.

The collapse result nevertheless exposes a limit in the predictive objective. Next-state prediction alone admits a trivial solution in which different observations receive nearly identical embeddings. SIGReg prevents that solution by forcing diversity across the latent space.

Diversity is not individuation. A well-spread vector space may still assign one entity several incompatible identities, merge different entities, or preserve no account of why an identity judgment holds. The regulariser ensures that the representation remains informative enough for prediction. It supplies no criterion governing what is one thing, what is two, or when the answer remains unresolved.

The last result is the one predictive architecture has nowhere to put. It can distribute probability over possible continuations. It cannot make unresolved identity binding on what happens next unless unresolved standing has been made a first-class condition outside the prediction itself.

The crossing

Put two identical cups on a table. Let one contain water and the other poison. Pass one behind a screen, cross their paths, and reveal a cup on the other side.

A good world model predicts trajectories through the occlusion. A better one maintains multiple hypotheses when the evidence is ambiguous. Both are doing epistemic work: estimating which represented continuation best explains what was seen.

The person about to drink faces a different question. Is the identity of this cup established strongly enough for the act to follow?

No increase in predictive fluency answers that question. A probability of 0.91 may be excellent tracking and an appalling basis for drinking. The threshold comes from the consequence, the governing criterion, and the authority under which the action is permitted. Those are not properties of the predicted trajectory.

The same structure appears in the river. A model may predict water at the bend from the terrain, season, cloud cover and every river it has ever seen. The direction carried an attribution: this river, this bend, this source, this interval, this standing. If the source confused two bends, if a flood moved the channel, or if the instruction has gone stale, the model's physical prediction and the instruction's authority separate. One may remain excellent while the other has failed.

The traveller needs the world model. The traveller also needs to know whether these words may still govern this walk.

LeCun's architecture addresses the first need and calls the result a route toward intelligence. The second need is logically prior wherever prediction becomes consequence.

Memory does not repair the order

The larger architecture proposed by LeCun includes memory. That matters, because a system acting through time must retain relevant state. Yet retained representation and persistent standing perform different work.

Memory carries information from an earlier computation into a later one. It may preserve an embedding, an observation, a predicted trajectory, a cost, a goal, or the internal state of the actor. Its success is measured by whether useful information remains available.

Persistence carries what remains the same, what changed, what is unresolved, which criteria produced that standing, and what may follow from it. Its success is measured by whether later consequence remains answerable to what was actually established.

Keeping an estimate longer does not turn it into identity. Rehearsing a claim does not renew its authority. Storing yesterday's direction does not make the river stay where it was.

This is why memory cannot be the foundation. A memory store can retain bits without individuating anything. To say that a memory held now is *of the same thing* encountered earlier invokes the continuity that memory was supposed to create.

The architecture has borrowed persistence before its memory module begins.

The order of dependence

LeCun's programme is organised around three powers: understanding, prediction and planning. Each arrives one step too late to serve as the ground.

Before a system can claim that an entity changed, it must establish which entity persists through the change.

Before it can treat a predicted state as the future of a particular entity, it must establish whose future the prediction concerns.

Before it can compare a predicted state with a goal, it must determine which criterion makes that state the goal's satisfaction.

Before it can plan an action, it must distinguish a physically available action from an admissible one.

Before it can learn from the result, it must preserve which action, entity, basis and unresolved conditions produced that result.

These are not extra capabilities to bolt onto a predictor after it becomes powerful enough. They establish the terms under which prediction counts as being about anything, and under which a plan may become an act.

The error is therefore architectural rather than empirical. Better video, longer temporal windows, hierarchical prediction, object-centred embeddings and more capable memory can improve the estimate at every stage. They cannot reverse the order of dependence. A representation may become astonishingly faithful while remaining unable to establish its own standing.

The world must stand before its changes can be predicted.

Back at the water

The robot reaches the bend. Its model has predicted the terrain, avoided the fallen branches and selected an efficient path. Every latent transition was excellent.

There is water where the instruction said it would be.

Now ask the question the model has never been required to answer. Is this the same bend? Did the instruction still bind at the time of arrival? Was its source entitled to direct this

person? Has anything occurred that requires the instruction to be revalidated? Does the action permitted by *find the river* extend to *drink from it*?

The physical world settles whether the water is there. A persistent frame settles what has been established about it. Governance settles what consequence may lawfully follow.

Prediction serves all three. It cannot substitute for any of them.

The world before the word

Return to the person who gave the directions.

They did something no predictor has yet been built to do, and they did it without effort, without training data, and without noticing. They made a claim about a standing world. They made it under criteria — this bend, not any bend. They attached it to a source, which was themselves, and which remains attached after they have gone. They let it bind across an interval they would not be present for. And they left it exposed: if the bend is dry, they were wrong, and the world says so.

That is the whole apparatus. It is ordinary. Every set of directions ever given contains it.

The deer at Prato showed what happens when one part of it goes unstated. The same animal, the same structure, two competent systems assigning different standing, and nobody able to say which criterion was in force because nobody had noticed there was a question. The unicorn showed what happens when the domain goes unbounded: a confident universal negative, held for centuries, running on evidence that never reached that far, waiting for something with a horn to walk out of the trees.

Neither is a failure of prediction. Prediction was never doing that work.

Language is not a system for encoding a world. It is the practice of addressing one that stands — picking something out, saying what holds of it, and remaining answerable for having said so. Everything else language does is built on that or borrowed from it. Prediction is one of the borrowings, and the most useful one we have.

The industry has spent a decade making the borrowing more fluent.

The lender was never specified.

Before the word, the world. Not as a metaphysical preference, and not as a condition of thought. As the thing that has to be there for a sentence to be about anything, for an instruction to still bind when the speaker is dead, for a claim to be capable of being wrong.

The river has to be there.

Everything else is what we say about it.